



Control Panel Sequence of Operations Altronic Control Panel Wellhead 2.0

Disclaimer

This Sequence of Operations (SOO) describes the general functionality and use of the Altronic, LLC control panel, including its operation with the DE-4000 Controller. This document is intended to provide a high-level overview of system behaviour and is not a substitute for proper engineering review, safety analysis, or site-specific implementation details.

Altronic, LLC makes no guarantees regarding the suitability, accuracy, or completeness of this SOO for any specific application. The DE-4000 Controller is a configurable system, and its operation depends on user-defined settings, logic, and parameters. As a result, Altronic, LLC cannot guarantee system performance under all configurations or applications. The use of this document and associated systems is at the sole risk of the user. Altronic, LLC is not responsible for any damages, malfunctions, or failures resulting from improper use, installation, modification, or misinterpretation of this document. Users should verify all control logic and safety protocols before commissioning.

This document is confidential and proprietary to Altronic, LLC. The information contained herein is intended for authorized personnel only and may not be reproduced, disclosed, or used in whole or in part without prior written consent from Altronic, LLC. Recipients agree to restrict access to this document to employees with a legitimate business need and to prevent any unauthorized disclosure. All contents of this document remain the exclusive property of Altronic, LLC and are subject to these confidentiality restrictions.

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0

Page 1 of 48

Revision History

[illegible]

Contents

System Overview5

DE-4000 Shutdown List.....6

System Start-Up Sequence 10

WELLHEAD / COMPRESSOR CONTROL SYSTEM20

1. SYSTEM DESCRIPTION20

2. OPERATING STATES.....20

3. WELLHEAD CONFIGURATION20

4. COMPRESSOR CONFIGURATION.....21

5. COMPRESSOR COMMUNICATION MONITORING21

6. COMPRESSOR DISCHARGE PRESSURE MONITORING21

7. COMPRESSOR DISCHARGE PRESSURE OVERRIDE22

8. PRIORITY MODE OPERATION (GAS / OIL)22

9. COMPRESSOR FLOW DISTRIBUTION.....23

10. TOTAL COMPRESSOR FLOW CAPACITY23

11. WELLHEAD ALLOCATION LOGIC24

12. WELL FLOW OFFSET CONTROL25

13. WELL FLOW STATUS MONITORING.....25

14. COMPRESSOR AVAILABILITY HANDLING26

15. COMMUNICATION LOSS OPERATION27

16. WELLHEAD PID CONTROL.....28

17. MANUAL / AUTOMATIC OPERATION.....28

18. NORMAL OPERATION SUMMARY29

19. SUCTION HEADER PRESSURE VALVE CONTROL29

19.1 DESCRIPTION.....29

19.2 CONFIGURABLE PARAMETERS29

19.3 SYSTEM MONITORING.....30

19.4 NORMAL OPERATION.....30

19.5 UPPER RANGE OPERATION30

19.6 HIGH PRESSURE CONDITION.....31

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0



19.7 COMPRESSOR SUCTION SETPOINT CALCULATION.....31

19.8 SUCTION HEADER VALVE CONTROL31

19.9 INTEGRATION WITH EXISTING CONTROL LOGIC31

20. STATION RECYCLE VALVE CONTROL32

20.1 DESCRIPTION.....32

20.2 CONFIGURATION32

20.3 CONTROL VARIABLE32

20.4 NORMAL OPERATION.....32

21. COMPRESSOR HMI CONTROL AND AUTO START LOGIC33

21.1 COMPRESSOR CONTROL PAGE.....33

21.2 MODE SELECTION33

21.3 AUTO START PERMISSION CONTROL33

21.4 AUTO START PERMISSION LOGIC.....34

21.5 REMOTE START COMMAND34

21.6 START CONFIRMATION MONITORING34

21.7 LOCAL VS REMOTE CONTROL INTERACTION35

21.8 INTEGRATION WITH SYSTEM OPTIMIZATION35

USER PARAMTERS AND SETTINGS36

ADDITIONAL MODBUS REGISTERS38

System Overview

The Altronic, LLC DE-4000 Series is a configurable Safety Shutdown and Control System designed to monitor, control, and protect critical rotating machinery. This microprocessor-based system processes both analog and digital inputs to execute automated start-up and shutdown sequences. Additional functions include pre-lube, engine warm-up, and cool-down operations.

The DE-4000 control panel includes a touchscreen HMI (Human-Machine Interface), which allows operators to start/stop equipment, adjust process setpoints, and monitor system status. Configuration of the DE-4000 is performed via the onboard programming interface, allowing users to tailor control sequences to specific application needs.

Support Documentation

This Sequence of Operations (SOO) is specific to the Altronic, LLC DE-4000 Control Panel and describes its operational functionality. For additional information, installation guidelines, configuration details, system drawings, and firmware updates, please refer to the following official documentation:

- DE-4000 Product Page: <https://www.altronic-llc.com/product/controls/de-4000/>
- DE-4000 Brochure: <https://www.altronic-llc.com/wp-content/uploads/DE-4000-Brochure.pdf>
- DE-4000 System Installation Instructions: <https://www.altronic-llc.com/documentation/de-4000-system-installation-instructions/>
- DE-4000 Operation and Configuration Manual: <https://www.altronic-llc.com/documentation/de-4000/operation-and-configuration-manual/>
- DE-4000 Firmware Download Quick Start Guide: <https://www.altronic-llc.com/wp-content/uploads/DE-4000-Firmware-Download-Quick-Start-Guide.pdf>
- System Drawings: (Refer to project-specific system drawings for wiring, connections, and installation details.)

Users should refer to these resources for detailed system specifications, installation procedures, system drawings, and firmware updates. This SOO document focuses specifically on the operation of the DE-4000 Control Panel and should be used in conjunction with official documentation to ensure proper configuration and usage.

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0

Page 5 of 48

DE-4000 Shutdown List

The DE-4000 Shutdown List provides a comprehensive overview of system fault conditions and their classifications. These faults are configured within the DE-4000 Control Panel to monitor critical parameters and trigger appropriate shutdown actions when necessary. The following tables detail these fault conditions, their settings, and response classifications.

Selectable Fault Types

Each fault in the DE-4000 system is categorized into one of the following fault classes, determining how and when the fault becomes active:

- A – Armed immediately
- B – Selectable time delay at reset before becoming armed
- C – Armed after the fault clears with a selectable time delay
- S – Scripted logic-based fault activation

These classifications ensure flexibility in system response, allowing operators to fine-tune safety and operational settings as needed.

	Channel	Description	I/O Type	Range	Unit of Measure	HIGH SD Class		HIGH Shutdown Setpoint	LOW or DI SD Class		LOW Shutdown Setpoint	HIGH Alarm Setpoint	LOW Alarm Setpoint	Modbus Reg 16 Bit (Node # set in DE4000)	Modbus Reg 32 Bit (Node 200)
						If - B/C	SECs		If - B	SECs					
Terminal Board 1	IN1	Wellhead Chocke #1 Feedback	4-20mA	0-100	%	N/A			N/A					40090	40001
	IN2	Wellhead Chocke #1 Feedback	4-20mA	0-100	%	N/A			N/A					40091	40003
	IN3	Wellhead Chocke #1 Feedback	4-20mA	0-100	%	N/A			N/A					40092	40005
	IN4	Wellhead Chocke #1 Feedback	4-20mA	0-100	%	N/A			N/A					40093	40007
	IN5	Wellhead Chocke #1 Feedback	4-20mA	0-100	%	N/A			N/A					40094	40009
	IN6	Wellhead Chocke #1 Feedback	4-20mA	0-100	%	N/A			N/A					40095	40011
	IN7	Wellhead Chocke #1 Feedback	4-20mA	0-100	%	N/A			N/A					40096	40013
	IN8	Wellhead Chocke #1 Feedback	4-20mA	0-100	%	N/A			N/A					40097	40015
	IN9	Wellhead Chocke #1 Feedback	4-20mA	0-100	%	N/A			N/A					40098	40017
	IN10	Wellhead Chocke #1 Feedback	4-20mA	0-100	%	N/A			N/A					40099	40019
	IN11	NOT USED												40100	40021
	IN12	NOT USED												40101	40023
	IN13	NOT USED												40102	40025
	IN14	NOT USED												40103	40027
	IN15	NOT USED												40104	40029
	IN16	NOT USED												40105	40031
	IN17	NOT USED												40106	40033
	IN18	NOT USED												40107	40035
	IN19	NOT USED												40108	40037
	IN20	NOT USED												40109	40039
	IN21	NOT USED												40110	40041
	IN22	NOT USED												40111	40043
	IN23	NOT USED												40112	40045
	IN24	NOT USED												40113	40047
	IN25	Run Status - Compressor #1	N/C DI						N/A					40114	40049
	IN26	Run Status - Compressor #2	N/C DI						N/A					40115	40051
	IN27	Run Status - Compressor #3	N/C DI						N/A					40116	40053
	IN28	Run Status - Compressor #4	N/C DI						N/A					40117	40055
	IN29	Remote Local Control	N/C DI						N/A					40118	40057
	IN30	Remote ESD From PLC	N/C DI						A					40119	40059
	IN31	NOT USED												40120	40061
	IN32	Local ESD	N/C DI						A					40121	40063
	AO1	Wellhead #1 Control												40260	-
	AO2	Wellhead #2 Control												40261	-
	AO3	Wellhead #3 Control												40262	-
	AO4	Wellhead #4 Control												40263	-
	DO1	Status Output												40020 Bit 0	-
	DO2													40020 Bit 1	-
	DO3													40020 Bit 2	-
	DO4													40020 Bit 3	-
	DO5													40020 Bit 4	-
	DO6													40020 Bit 5	-
	DO7													40020 Bit 6	-
	DO8													40020 Bit 7	-
	RPM1													40250	-
	RPM2													40251	-

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0

Terminal Board 2	Channel	Description	I/O Type	Range	Unit of Measure	HIGH SD Class (If - B) SECs		HIGH Shutdown Setpoint	LOW or DI SD Class (If - B) SECs		LOW Shutdown Setpoint	HIGH Alarm Setpoint	LOW Alarm Setpoint	Modbus Reg 16 Bit (Node # set in DE4000)	Modbus Reg 32 Bit (Node 200)
	IN1		NOT USED											40122	40065
	IN2		NOT USED											40123	40067
	IN3		NOT USED											40124	40069
	IN4		NOT USED											40125	40071
	IN5		NOT USED											40126	40073
	IN6		NOT USED											40127	40075
	IN7		NOT USED											40128	40077
	IN8		NOT USED											40129	40079
	IN9		NOT USED											40130	40081
	IN10		NOT USED											40131	40083
	IN11		NOT USED											40132	40085
	IN12		NOT USED											40133	40087
	IN13		NOT USED											40134	40089
	IN14		NOT USED											40135	40091
	IN15		NOT USED											40136	40093
	IN16		NOT USED											40137	40095
	IN17		NOT USED											40138	40097
	IN18		NOT USED											40139	40099
	IN19		NOT USED											40140	40101
	IN20		NOT USED											40141	40103
	IN21		NOT USED											40142	40105
	IN22		NOT USED											40143	40107
	IN23		NOT USED											40144	40109
	IN24		NOT USED											40145	40111
	IN25		NOT USED											40146	40113
	IN26		NOT USED											40147	40115
	IN27		NOT USED											40148	40117
	IN28		NOT USED											40149	40119
	IN29		NOT USED											40150	40121
	IN30		NOT USED											40151	40123
	IN31		NOT USED											40152	40125
IN32		NOT USED											40153	40127	
AO1	Wellhead #5 Control													40264	-
AO2	Wellhead #6 Control													40265	-
AO3	Wellhead #7 Control													40266	-
AO4	Wellhead #8 Control													40267	-
DO1														40021 Bit 0	-
DO2														40021 Bit 1	-
DO3														40021 Bit 2	-
DO4														40021 Bit 3	-
DO5														40021 Bit 4	-
DO6														40021 Bit 5	-
DO7														40021 Bit 6	-
DO8														40021 Bit 7	-
RPM1														40252	-
RPM2														40253	-

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0

Channel	Description	I/O Type	Range	Unit of Measure	HIGH		HIGH Shutdown Setpoint	LOW or DI		LOW Shutdown Setpoint	HIGH Alarm Setpoint	LOW Alarm Setpoint	Modbus Reg 16 Bit (Node # set in DE4000)	Modbus Reg 32 Bit (Node 200)
					SDClass	(If - B) SECs		SDClass	(If - B) SECs					
IN1		NOT USED											40154	40129
IN2		NOT USED											40155	40131
IN3		NOT USED											40156	40133
IN4		NOT USED											40157	40135
IN5		NOT USED											40158	40137
IN6		NOT USED											40159	40139
IN7		NOT USED											40160	40141
IN8		NOT USED											40161	40143
IN9		NOT USED											40162	40145
IN10		NOT USED											40163	40147
IN11		NOT USED											40164	40149
IN12		NOT USED											40165	40151
IN13		NOT USED											40166	40153
IN14		NOT USED											40167	40155
IN15		NOT USED											40168	40157
IN16		NOT USED											40169	40159
IN17		NOT USED											40170	40161
IN18		NOT USED											40171	40163
IN19		NOT USED											40172	40165
IN20		NOT USED											40173	40167
IN21		NOT USED											40174	40169
IN22		NOT USED											40175	40171
IN23		NOT USED											40176	40173
IN24		NOT USED											40177	40175
IN25		NOT USED											40178	40177
IN26		NOT USED											40179	40179
IN27		NOT USED											40180	40181
IN28		NOT USED											40181	40183
IN29		NOT USED											40182	40185
IN30		NOT USED											40183	40187
IN31		NOT USED											40184	40189
IN32		NOT USED											40185	40191
AO1	Wellhead #9 Control												40272	-
AO2	Wellhead #10 Control												40273	-
AO3	Suction Header Valve												40274	-
AO4	Station Recycle Valve												40275	-
DO1													40022 Bit 0	-
DO2													40022 Bit 1	-
DO3													40022 Bit 2	-
DO4													40022 Bit 3	-
DO5													40022 Bit 4	-
DO6													40022 Bit 5	-
DO7													40022 Bit 6	-
DO8													40022 Bit 7	-
RPM1													40254	-
RPM2													40255	-

Terminal Board 3

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0

Page 9 of 48

System Start-Up Sequence

The Start-Up menu allows users to configure an auto start sequence with a high degree of flexibility. The system progresses through ten sequential states (State 0 to State 9), with transitions controlled by configured permissive and timers. The Start Button initiates the sequence.

Only the DS Outputs are configurable within the Start-Up Sequence. The Terminal Board Digital Outputs (DO) should be configured separately via the Outputs page or through scripting. Refer to the DE-4000 Operation and Configuration Manual for configuring Digital Outputs. The setup and control of DOs within this system can be found in the Digital Outputs section of this document.

To bypass a state, set its timers to 0 and disable the permissive by unchecking the Enable checkbox.

For specific details, including timing parameters and configuration options, refer to the DE-4000 Operation and Configuration Manual.

State 0 Ready to Start

Initial state. When all faults are cleared, the Start Button can be pressed to begin the sequence.

Additional Notes
DS1 = Enable To Murphy Units Compressor 1 = De-energized
DS2 = Enable To Murphy Units Compressor 2 = De-energized
DS3 = Enable To Murphy Units Compressor 3 = De-energized
DS4 = Enable To Murphy Units Compressor 4 = De-energized

State 1

Label	Not Used	Timer:		0	min	0	sec
		Timeout:		0	min	0	sec
DS Outputs							
Output On (Beginning of State)	☐	DS1	☐	DS2	☐	DS3	DS4
Output Off (End of State)	☐	DS1	☐	DS2	☐	DS3	DS4
Permissive A							
☐	Enable	Channel				REL	Setpoint
		Channel Name:					
Permissive B							
☐	Enable	Channel				REL	Setpoint
		Channel Name:					
Additional Notes							

State 2

Label	Not Used	Timer:		0	min	0	sec
		Timeout:		0	min	0	sec
DS Outputs							
Output On (Beginning of State)	☐	DS1	☐	DS2	☐	DS3	DS4
Output Off (End of State)	☐	DS1	☐	DS2	☐	DS3	DS4
Permissive A							
☐	Enable	Channel				REL	Setpoint
		Channel Name:					
Permissive B							
☐	Enable	Channel				REL	Setpoint
		Channel Name:					
Additional Notes							

State 3

Label	Not Used	Timer:		0	min	0	sec
		Timeout:		0	min	0	sec
DS Outputs							
Output On (Beginning of State)	☐	DS1	☐	DS2	☐	DS3	DS4
Output Off (End of State)	☐	DS1	☐	DS2	☐	DS3	DS4
Permissive A							
☐	Enable	Channel				REL	Setpoint
		Channel Name:					
Permissive B							
☐	Enable	Channel				REL	Setpoint
		Channel Name:					
Additional Notes							

State 4

Label	Not Used	Timer:		0	min	0	sec
		Timeout:		0	min	0	sec
DS Outputs							
Output On (Beginning of State)	☐	DS1	☐	DS2	☐	DS3	DS4
Output Off (End of State)	☐	DS1	☐	DS2	☐	DS3	DS4
Permissive A							
☐	Enable	Channel				REL	Setpoint
		Channel Name:					
Permissive B							
☐	Enable	Channel				REL	Setpoint
		Channel Name:					
Additional Notes							

State 5

Label	Not Used	Timer:		0	min	0	sec
		Timeout:		0	min	0	sec
DS Outputs							
Output On (Beginning of State)	☐	DS1	☐	DS2	☐	DS3	DS4
Output Off (End of State)	☐	DS1	☐	DS2	☐	DS3	DS4
Permissive A							
☐	Enable	Channel				REL	Setpoint
		Channel Name:					
Permissive B							
☐	Enable	Channel				REL	Setpoint
		Channel Name:					
Additional Notes							

State 6

Label	Not Used	Timer:		0	min	0	sec
		Timeout:		0	min	0	sec
DS Outputs							
Output On (Beginning of State)	☐	DS1	☐	DS2	☐	DS3	DS4
Output Off (End of State)	☐	DS1	☐	DS2	☐	DS3	DS4
Permissive A							
☐	Enable	Channel				REL	Setpoint
		Channel Name:					
Permissive B							
☐	Enable	Channel				REL	Setpoint
		Channel Name:					
Additional Notes							

State 7

Label	Not Used	Timer:		0	min	0	sec
		Timeout:		0	min	0	sec
DS Outputs							
Output On (Beginning of State)	☐	DS1	☐	DS2	☐	DS3	DS4
Output Off (End of State)	☐	DS1	☐	DS2	☐	DS3	DS4
Permissive A							
☐	Enable	Channel				REL	Setpoint
		Channel Name:					
Permissive B							
☐	Enable	Channel				REL	Setpoint
		Channel Name:					
Additional Notes							

State 8

Label	Running							
DS Outputs								
Output On (Beginning of State)	☒	DS1	☒	DS2	☒	DS3	☒	DS4
Output Off (End of State)	☐	DS1	☐	DS2	☐	DS3	☐	DS4
Additional Notes								

State 9

Label	Not Used	Timer:		0	min	0	sec
		Timeout:		0	min	0	sec
DS Outputs							
Output On (Beginning of State)	☐	DS1	☐	DS2	☐	DS3	DS4
Output Off (End of State)	☐	DS1	☐	DS2	☐	DS3	DS4
Permissive A							
☐	Enable	Channel				REL	Setpoint
		Channel Name:					
Permissive B							
☐	Enable	Channel				REL	Setpoint
		Channel Name:					
Additional Notes							

1. SYSTEM DESCRIPTION

The Wellhead and Compressor Control System automatically coordinates gas flow from multiple wellheads through a shared compressor system. The control strategy dynamically allocates compressor capacity to individual wells based on operator-defined priorities, real-time operating conditions, and system protection limits. The system is designed to maximize production while preventing compressor overload, excessive discharge pressure, and unstable operation.

2. OPERATING STATES

The control system operates based on the following states:

- **State < 8 – Startup / Idle:**

Automatic well and compressor coordination is disabled. All offsets, timers, and priority logic are reset.

- **State ≥ 8 – Automatic Operation:**

The system actively controls compressor flow allocation, well enable status, and flow setpoints.

3. WELLHEAD CONFIGURATION

Each wellhead is configured with the following parameters:

- Operator-entered maximum flow (HMI Desired Flow)
- Calculated desired flow setpoint generated by the control logic
- Actual injection flow feedback
- Gas priority ranking
- Oil priority ranking
- Automatic enable/disable status

Wellheads are automatically enabled or disabled based on available compressor capacity and the active priority mode.

4. COMPRESSOR CONFIGURATION

Each compressor includes:

- Maximum rated flow capacity
- Current measured flow
- Desired flow setpoint
- Run status
- Discharge pressure measurement and setpoint
- Suction pressure measurement and setpoint

Only compressors that are running are considered available for flow allocation.

5. COMPRESSOR COMMUNICATION MONITORING

When heartbeat monitoring is enabled:

- Each running compressor is continuously monitored for communication loss
 - If communication is lost for longer than the configured fault delay, a compressor fault is generated
 - Faulted compressors are excluded from all flow calculations and allocation logic
-

6. COMPRESSOR DISCHARGE PRESSURE MONITORING

The system continuously monitors compressor discharge pressure relative to the configured setpoint.

- A high-pressure condition is identified when:
 - Discharge Pressure > Setpoint + Deadband %
 - Only compressors that are running and not in communication fault are evaluated
-

7. COMPRESSOR DISCHARGE PRESSURE OVERRIDE

If any compressor exceeds the discharge pressure limit:

- The system activates a global discharge pressure override
- An incremental flow adder is applied to all enabled wells
- The adder increases over time based on a configured interval

This action increases total well flow to reduce compressor discharge pressure.

When the condition clears:

- The override adder is reset to its baseline value
 - Normal flow allocation resumes
-

8. PRIORITY MODE OPERATION (GAS / OIL)

The system operates in either Gas Priority Mode or Oil Priority Mode.

- **Default Mode:** Gas Priority
 - The system switches to Oil Priority when:
 - The number of running wells meets or exceeds a configured threshold
 - A stabilization timer has expired
 - If conditions are no longer met:
 - The system reverts back to Gas Priority
 - A countdown timer may be displayed during the transition period
-

9. COMPRESSOR FLOW DISTRIBUTION

The system calculates and distributes required flow across all running compressors.

- Total required flow is calculated as the sum of all well desired flows
- Flow is distributed proportionally based on compressor capacity:

$(\text{Compressor Max Flow} / \text{Total Available Capacity}) \times \text{Total Required Flow}$

- Each compressor's assigned flow is limited to its maximum capacity
 - Compressors that are not running are assigned zero flow
-

10. TOTAL COMPRESSOR FLOW CAPACITY

The total available compressor flow is calculated using real-time data:

- Actual compressor flow is used when within limits
- If actual flow exceeds maximum capacity plus deadband:
 - Flow is capped at maximum rated capacity

This total defines the available flow that can be allocated to wellheads.

11. WELLHEAD ALLOCATION LOGIC

Wellheads are enabled and assigned flow based on priority and available compressor capacity.

Priority Sorting:

- Wells are sorted by Gas Priority or Oil Priority depending on active mode
 - Lower priority value indicates higher importance
-

Flow Allocation:

- Wells are enabled sequentially based on priority
 - Each enabled well is assigned flow up to its configured maximum
 - Allocation continues until available compressor capacity is fully utilized
 - If insufficient capacity exists:
 - Lower-priority wells are disabled
-

Override Condition:

- When discharge pressure override is active:
 - Additional flow is added to enabled wells only
-

12. WELL FLOW OFFSET CONTROL

The system continuously adjusts well flow to match compressor capacity.

- Target condition:

Total Well Flow $\approx$ Total Compressor Capacity

Adjustment Logic:

After a stabilization period:

- If actual well flow is below target:
 - Offset is decreased to increase flow demand
 - If actual well flow is above target:
 - Offset is increased to reduce flow demand
-

Reset Conditions:

Offset is reset when:

- Compressor discharge override is active
 - Any well is in manual mode
 - Number of active wells changes
 - Number of running compressors changes
-

13. WELL FLOW STATUS MONITORING

Each enabled well is evaluated to confirm proper flow performance.

- A minimum acceptable flow is calculated using a configurable deadband
 - If actual flow $\geq$ minimum threshold:
 - Flow status is set to GOOD
 - If actual flow $<$ minimum threshold:
 - Flow status is set to BAD
-

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0

Page 25 of 48

14. COMPRESSOR AVAILABILITY HANDLING

If no compressors are running:

- All well flow control is suspended
 - Wells are effectively disabled
 - System displays a status message indicating compressors are offline
-

15. COMMUNICATION LOSS OPERATION

- When a communication loss condition is detected on the Wellhead PLC:
 - The system enters a communication override mode
 - A system status message is displayed indicating communication/network loss
 - Communication Loss Override logic becomes active and takes priority over normal PID control
- Wellhead outputs are driven according to the configured override type (Preset or Adaptive)
 - If Override Type = Preset, each wellhead output is set to a fixed, configurable percentage
 - If Override Type = Hold + Adder, each wellhead output is based on the last known PID output and is gradually increased over time using a separate discharge adder, up to a maximum of 100%
 - The last known PID outputs are captured at the time communication is lost and used as reference values during override operation
- **Compressor Communication Loss Handling:**
 - Loss of communication with compressor PLCs shall **not** trigger Communication Override mode
 - Compressor communication loss shall be annunciated on the HMI banner for operator awareness only
 - Normal wellhead control strategy shall continue unless a separate protective condition is triggered
- Upon restoration of communication, override mode is cleared, reference values are updated, and normal PID control resumes
- If the system is not in a valid running state or compressors are offline, all wellhead outputs are forced to 0% for safe operation
- If the chokes, sales valves, and recycle valves are controlled via communications, no corrective action shall be taken during a loss of communication

Available Modes:

- **Fixed Output Mode:**

Wells are driven to a predefined output value

- **Hold/Adder Output Mode:**

Well outputs increase gradually over time based on a configured rate

16. WELLHEAD PID CONTROL

Each wellhead operates under PID control in Automatic Mode.

- PID setpoint is equal to the calculated desired flow

Control Behavior:

- If the well is enabled:
 - PID operates normally once minimum flow threshold is met
 - Below threshold, a minimum output is enforced
- If the well is disabled:
 - Output is forced to zero
- If compressors are offline:
 - All PID outputs are forced to zero

17. MANUAL / AUTOMATIC OPERATION

Each wellhead supports both manual and automatic control modes.

- **Automatic Mode:**
 - System controls flow and allocation
- **Manual Mode:**
 - Operator directly controls output
 - Automatic offset adjustments are disabled
- Remote control selection forces all wells into Automatic Mode

18. NORMAL OPERATION SUMMARY

- Compressors determine available system capacity
 - Wells are enabled based on priority and demand
 - Flow is distributed proportionally across compressors
 - Offset logic maintains flow balance
 - Override logic protects compressors from high pressure
 - Communication loss handling ensures safe fallback operation
-

19. SUCTION HEADER PRESSURE VALVE CONTROL

19.1 DESCRIPTION

The system shall monitor and regulate the suction header pressure to maintain stable operation of the compressor system and optimize gas flow handling.

The Pad Optimization Panel shall control suction header pressure relative to a configurable target using available system resources, including compressor coordination and optional suction header valve control.

19.2 CONFIGURABLE PARAMETERS

The following parameters shall be available within the Pad Optimization Panel:

- Suction Header Target Pressure (SCADA or operator input)
 - Suction Header High Range Setting (Upper Range)
 - Min Suction Header Pressure Setting
 - Compressor Stagger Offset (psi) – Default: 0
 - Suction Header Valve Enable (Enable / Disable)
-

19.3 SYSTEM MONITORING

The system shall continuously monitor:

- Suction Header Pressure
 - Compressor run status
-

19.4 NORMAL OPERATION

During normal operation, the system shall maintain:

Suction Header Pressure $\approx$ Suction Header Target Pressure

Control actions may include:

- Compressor loading / unloading coordination
 - Flow allocation adjustments
-

19.5 UPPER RANGE OPERATION

The Suction Header High Range defines the allowable pressure deviation above the target.

The system may allow suction pressure to rise to:

Target Pressure + High Range

This condition may occur during:

- Compressor restart events (temporary setpoint adjustment for smooth recovery)
- Well unload events
- Temporary gas supply imbalance
- Suction header pre-pack for anticipated demand

If compressors are not running, suction pressure may increase up to the High Range limit without corrective action.

19.6 HIGH PRESSURE CONDITION

A high suction pressure condition is defined as:

$\text{Suction Pressure} > (\text{Target Pressure} + \text{High Range})$

When this condition occurs, the Pad Optimization Panel may initiate corrective actions, including:

- Compressor staggering using configured offset values
- Activation or modulation of suction header valve (if enabled)

These actions are intended to reduce suction pressure and stabilize system operation.

19.7 COMPRESSOR SUCTION SETPOINT CALCULATION

The system shall calculate the automatic suction pressure setpoint for compressors as follows:

$\text{Compressor Auto Suction SP} = \text{Min Suction Header Pressure Setting}$

This value shall be written to all applicable compressors via Modbus communication.

19.8 SUCTION HEADER VALVE CONTROL

When enabled, the suction header valve shall be used as an additional control element to regulate suction header pressure.

- Valve output shall be controlled via analog signal (4–20 mA) or over Modbus communications
 - Valve action shall occur only during high suction pressure conditions
 - Valve shall remain inactive when disabled
-

19.9 INTEGRATION WITH EXISTING CONTROL LOGIC

This suction pressure control strategy operates in parallel with:

- Wellhead flow allocation logic
- Compressor flow distribution
- Discharge pressure override logic
- Communication loss override logic

The system shall prioritize stable compressor operation while maintaining suction pressure within the defined operating range.

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0

Page 31 of 48

20. STATION RECYCLE VALVE CONTROL (CLOSED LOOP PID)

20.1 DESCRIPTION

The station may be equipped with a single recycle (bypass) valve used to maintain stable compressor operation during low-flow or transient conditions.

The recycle valve shall operate under closed-loop PID control using a default control strategy independent of well allocation logic.

20.2 CONFIGURATION

The following parameters shall be configurable:

- Recycle Valve Enable (Enable / Disable)
 - PID Setpoint
 - PID Tuning Parameters (Proportional, Integral, Derivative)
 - Minimum Valve Position (%)
 - Maximum Valve Position (%)
-

20.3 CONTROL VARIABLE

The recycle valve PID shall control based on a variable.

- Station Header Pressure

The selected control variable and setpoint shall be defined during commissioning.

20.4 NORMAL OPERATION

When enabled:

- The recycle valve shall modulate via PID control to maintain the configured setpoint
- Valve position shall be constrained within configured minimum and maximum limits
- The control loop shall operate continuously to stabilize compressor operation

21. COMPRESSOR HMI CONTROL AND AUTO START LOGIC

21.1 COMPRESSOR CONTROL PAGE

The Pad Optimization Panel shall include a dedicated Compressor Control Page for operator interaction.

For each compressor, the following shall be provided on the HMI:

- Compressor Status
 - Start Command Button
 - Stop Command Button
 - Mode Selector
-

21.2 MODE SELECTION

Each compressor shall support the following operating modes:

Manual Mode:

- Operator has full control of compressor operation
- Start and Stop commands are issued directly from the HMI
- Pad Optimization Panel does not issue automatic start commands

Auto Mode:

- Compressor operation is controlled by the Pad Optimization Panel
 - Start permission and commands are issued based on system demand and optimization logic
-

21.3 AUTO START PERMISSION CONTROL

The Pad Optimization Panel shall control compressor auto start permission via Modbus communication.

Register:

- Auto Start Allowed

Values:

- 0 = Auto Start Not Allowed
 - 1 = Auto Start Allowed
-

21.4 AUTO START PERMISSION LOGIC

The system shall manage Auto Start Allowed as follows:

- If compressor status indicates shutdown or communication loss:
 - The panel shall write:

Auto Start Allowed = 0

- The panel shall not write **Auto Start Allowed = 1** until:
 - The optimizer determines that additional compressor capacity is required
- After any reset or shutdown condition:
 - The system shall wait a minimum of 10 seconds before re-evaluating compressor status

21.5 REMOTE START COMMAND

When the optimizer determines that a compressor is required:

1. The panel shall write:
Auto Start Allowed = 1
2. The panel shall issue a command to the Remote Start Register

This command shall function as a reset/start signal to the compressor control system (Murphy panel).

21.6 START CONFIRMATION MONITORING

After issuing a remote start command, the system shall verify compressor response.

- The panel shall monitor the **Murphy Skid State Register**
- If the compressor does not transition into a start or running state:
 - The panel shall reissue the Remote Start Command every 5 minutes
- This retry sequence shall continue until:
 - The compressor state changes to indicate successful start
 - Or the optimizer removes the start request

21.7 LOCAL VS REMOTE CONTROL INTERACTION

- In **Manual Mode**, operator commands take priority and no remote start commands shall be issued
 - In **Auto Mode**, the Pad Optimization Panel has full control of:
 - Start permission
 - Start command control
 - Start retry logic
 - Mode transitions shall not automatically start or stop compressors unless required by system conditions
-

21.8 INTEGRATION WITH SYSTEM OPTIMIZATION

Compressor auto start logic shall operate in coordination with:

- Total compressor flow demand
- Wellhead allocation logic
- Suction header pressure control (Section 19)
- Compressor protection and override conditions

The system shall ensure compressors are only started when required to meet system demand while maintaining stable operation.

USER PARAMETERS AND SETTINGS

FUXA & Dashboard Settings			
Parameter	Default	Description	SOO Behavior
iframe_page	5	Enter Dashboard Page Number for FUXA Flow View	The HMI or SCADA system loads the specified page number in the FUXA dashboard to display wellhead flow data.

System Configuration			
Parameter	Default	Description	SOO Behavior
WellHead_Priority_Release	4	Well number to switch from gas to oil priority	When gas-to-oil priority is required, the system automatically shifts control focus to the well number specified here.
Flow_Offset_SP	0.001	Amount to adjust compressor flow to increase wellhead flow	Small incremental changes are applied to compressor flow setpoints to optimize distribution to wellheads.
Flow_Offset_SP_Max_Offset	0.05	Maximum flow adjustment to even out wellhead distribution	Limits total adjustment to prevent excessive flow swings.
Compressor_Flow_Max_Deadband	0.1	Deadband for actual vs max compressor flow	Prevents oscillations by ignoring minor deviations between measured and max flow.
FLOW_Status_Mode	0	0 = Desired Flow, 1 = Calculated Flow	Determines whether wellhead flow monitoring uses operator-set desired flow or system-calculated flow.
Compressor_DSCH_Prs_SP	800	Expected discharge pressure for the system	Target discharge pressure for all compressors.
Compressor_DSCH_Prs_Deadband	3	Deadband for discharge override	Prevents immediate override for minor fluctuations above/below discharge pressure.
Offset_Adjustment_ZEROTIER	0.006	Flow sensor offset correction	Adjusts sensor readings to align with actual flow if calibration required.
Desired_WellFlow_DSCHOverride_DB	0.1	Max adjustment for well flow during discharge override	Limits adjustment applied to well setpoints during high discharge pressure.
Desired_WellFlow_DSCHOverride_Adder	0.01	Incremental adjustment value per interval	Step value used to adjust each well's flow setpoint during override.
Desired_WellFlow_DSCHOverride_Time	5 sec	Time interval between well setpoint adjustments	System updates well flow setpoints periodically when in discharge override mode.

Modbus Configuration			
Parameter	Default	Description	SOO Behavior
Compressor_1_Name – Compressor_4_Name	Comp1 – Comp4	AWI names for compressors	Names used for Modbus communication; spaces are replaced with dashes.
WellHead_PLC_Name	WellHead-Scada	AWI name for wellhead SCADA	Sets PLC name for Modbus communications.

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0

Page 36 of 48

ALTRONIC

Wellhead PID Configuration			
Parameter	Default	Description	SOO Behavior
WellHead_##_PID_Min	50	Minimum PID output	Ensures valves do not close below safe minimum.
WellHead_##_PID_Flow_Release	0.1	Flow to release PID	Flow threshold to engage PID control.
WellHead_##_Flow_Status_Percent_Deadband	3%	Deadband for flow status monitoring	Small deviations from setpoint do not trigger corrective action.

Communication Override Settings			
Parameter	Default	Description	SOO Behavior
WellHead_Comms_Override_Type	0	0 = Presets, 1 = Current Position Hold	Determines valve behavior when communication is lost.
WellHead_Comms_Debounce	2 sec	Debounce time for comms loss	Prevents false triggers from momentary communication glitches.
Comms_Lose_DSCH_Override_Adder	2%	Percent valve opening per time step	Gradually opens valve if communication is lost during high discharge.
Comms_Lose_DSCH_Override_Adder_Time	5 sec	Time interval between increments	Applies adder stepwise until preset position reached.
WellHead_##_Comms_Lose_Preset_Percent	50%	PID output or valve position during comms loss	Sets safe fallback valve position for each well.

Suction Header Valve Settings			
Parameter	Default	Description	SOO Behavior
Suction_Header_Valve_Enable	0	0 = Disabled, 1 = Enabled	Determines whether valve participates in control strategy
Suction_Header_Target_Pressure	20 psi	Target pressure for suction header valve PID.	Used as primary control setpoint for system optimization
Suction_Header_Valve_Max_Prs_SP	25 psi	Debounce time for comms loss	Defines high-end limit for suction pressure control
Suction_Header_Valve_Min_Prs_SP	18 psi	Percent valve opening per time step	Prevents system from operating below safe pressure
Compressor_Starting_Staggering_Offset_PSI	0 psi	The PSI pressure to add to target to start next compressor	Determines whether valve participates in control strategy

Recycle Valve Settings			
Parameter	Default	Description	SOO Behavior
Recycle_Valve_Max_Pos	100 %	Maximum allowable valve position (%)	Limits valve from opening beyond design capacity
Recycle_Valve_Min_Pos	0 %	Minimum allowable valve position (%)	Prevents valve from closing below safe operating limit

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0

Page 37 of 48

ADDITIONAL MODBUS REGISTERS

Register	Description(FROM DE TO FUXA / MLink)	Notes
460000	Number Of Configured Terminal Boards	
460002	Compressor #1 Desire Flow SP For PID Murphy	
460004	Compressor #2 Desire Flow SP For PID Murphy	
460006	Compressor #3 Desire Flow SP For PID Murphy	
460008	Compressor #4 Desire Flow SP For PID Murphy	
460010	Reserved For Future	
460012	Offset Value	
460014	Calculated Flow With Offset	
460016	Priority Mode	0 = Gas, 1 = Oil
460018	Wellhead Control In Override	DSCH > Parameter
460020	Wellhead Control In Override Based Off Comp Speed SP	DSCH > Parameter, SUCT < Parameter
460022	Reserved For Future	
460024	Reserved For Future	
460026	Wellhead #1 In Manual/Auto	0 = Auto, 1 = Manual
460028	Wellhead #2 In Manual/Auto	0 = Auto, 1 = Manual
460030	Wellhead #3 In Manual/Auto	0 = Auto, 1 = Manual
460032	Wellhead #4 In Manual/Auto	0 = Auto, 1 = Manual
460034	Wellhead #5 In Manual/Auto	0 = Auto, 1 = Manual
460036	Wellhead #6 In Manual/Auto	0 = Auto, 1 = Manual
460038	Wellhead #7 In Manual/Auto	0 = Auto, 1 = Manual
460040	Wellhead #8 In Manual/Auto	0 = Auto, 1 = Manual
460042	Wellhead #9 In Manual/Auto	0 = Auto, 1 = Manual
460044	Wellhead #10 In Manual/Auto	0 = Auto, 1 = Manual
460046	Reserved For Future	
460048	Reserved For Future	
460050	Wellhead #1 Calculated Desired Flow	
460052	Wellhead #2 Calculated Desired Flow	
460054	Wellhead #3 Calculated Desired Flow	
460056	Wellhead #4 Calculated Desired Flow	
460058	Wellhead #5 Calculated Desired Flow	
460060	Wellhead #6 Calculated Desired Flow	
460062	Wellhead #7 Calculated Desired Flow	
460064	Wellhead #8 Calculated Desired Flow	
460066	Wellhead #9 Calculated Desired Flow	
460068	Wellhead #10 Calculated Desired Flow	
460070	Reserved For Future	
460072	Reserved For Future	
460074	Wellhead #1 Running Status	0 = NOT RUNNING, 1 = Running

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0

Page 38 of 48

ALTRONIC

460076	Wellhead #2 Running Status	0 = NOT RUNNING, 1 = Running
460078	Wellhead #3 Running Status	0 = NOT RUNNING, 1 = Running
460080	Wellhead #4 Running Status	0 = NOT RUNNING, 1 = Running
460082	Wellhead #5 Running Status	0 = NOT RUNNING, 1 = Running
460084	Wellhead #6 Running Status	0 = NOT RUNNING, 1 = Running
460086	Wellhead #7 Running Status	0 = NOT RUNNING, 1 = Running
460088	Wellhead #8 Running Status	0 = NOT RUNNING, 1 = Running
460090	Wellhead #9 Running Status	0 = NOT RUNNING, 1 = Running
460092	Wellhead #10 Running Status	0 = NOT RUNNING, 1 = Running
460094	Reserved For Future	
460096	Reserved For Future	
460098	Reserved For Future	
460100	Wellhead #1 Flow Running Status	0 = No Flow, 1 = Flow Is Within Range
460102	Wellhead #2 Flow Running Status	0 = No Flow, 1 = Flow Is Within Range
460104	Wellhead #3 Flow Running Status	0 = No Flow, 1 = Flow Is Within Range
460106	Wellhead #4 Flow Running Status	0 = No Flow, 1 = Flow Is Within Range
460108	Wellhead #5 Flow Running Status	0 = No Flow, 1 = Flow Is Within Range
460110	Wellhead #6 Flow Running Status	0 = No Flow, 1 = Flow Is Within Range
460112	Wellhead #7 Flow Running Status	0 = No Flow, 1 = Flow Is Within Range
460114	Wellhead #8 Flow Running Status	0 = No Flow, 1 = Flow Is Within Range
460116	Wellhead #9 Flow Running Status	0 = No Flow, 1 = Flow Is Within Range
460118	Wellhead #10 Flow Running Status	0 = No Flow, 1 = Flow Is Within Range
460120	Reserved For Future	
460122	Reserved For Future	
460124	Reserved For Future	
460126	Reserved For Future	
460128	Reserved For Future	
460130	iFrame Dashboard Page	Number relates to which dashboard page Fuxa will display on
460132	Heartbeat Fault Enable	For Fuxa And PLC. 0 = Disabled, 1 = Enabled
460134	Wellhead Priority Release	When To Switch From Gas To Oil Priority
460136	Flow Offset Setpoint	Offset Setpoint When Adjusting Flow on Wells

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0

Page 39 of 48

460138	Compressor Flow Max Deadband	How Much Actual Flow Above Max Flow Allowed For Calcs
460140	Flow Status Mode	0 = Desired Flow, 1 = Calculated Flow
460142	Wellhead #1 PID Min Output	When Running But Injection Is Less Than Flow Release
460144	Wellhead #1 PID Flow Release	When Injection Rate Flow Is Enough To Run Calcs
460146	Wellhead #1 Flow Running Status Percent	What Percent of Desired Flow To Say Well Is Flowing For Status
460148	Wellhead #2 PID Min Output	When Running But Injection Is Less Than Flow Release
460150	Wellhead #2 PID Flow Release	When Injection Rate Flow Is Enough To Run Calcs
460152	Wellhead #2 Flow Running Status Percent	What Percent of Desired Flow To Say Well Is Flowing For Status
460154	Wellhead #3 PID Min Output	When Running But Injection Is Less Than Flow Release
460156	Wellhead #3 PID Flow Release	When Injection Rate Flow Is Enough To Run Calcs
460158	Wellhead #3 Flow Running Status Percent	What Percent of Desired Flow To Say Well Is Flowing For Status
460160	Wellhead #4 PID Min Output	When Running But Injection Is Less Than Flow Release
460162	Wellhead #4 PID Flow Release	When Injection Rate Flow Is Enough To Run Calcs
460164	Wellhead #4 Flow Running Status Percent	What Percent of Desired Flow To Say Well Is Flowing For Status
460166	Wellhead #5 PID Min Output	When Running But Injection Is Less Than Flow Release
460168	Wellhead #5 PID Flow Release	When Injection Rate Flow Is Enough To Run Calcs
460170	Wellhead #5 Flow Running Status Percent	What Percent of Desired Flow To Say Well Is Flowing For Status
460172	Wellhead #6 PID Min Output	When Running But Injection Is Less Than Flow Release
460174	Wellhead #6 PID Flow Release	When Injection Rate Flow Is Enough To Run Calcs
460176	Wellhead #6 Flow Running Status Percent	What Percent of Desired Flow To Say Well Is Flowing For Status
460178	Wellhead #7 PID Min Output	When Running But Injection Is Less Than Flow Release
460180	Wellhead #7 PID Flow Release	When Injection Rate Flow Is Enough To Run Calcs
460182	Wellhead #7 Flow Running Status Percent	What Percent of Desired Flow To Say Well Is Flowing For Status

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0

Page 40 of 48

ALTRONIC

460184	Wellhead #8 PID Min Output	When Running But Injection Is Less Then Flow Release
460186	Wellhead #8 PID Flow Release	Wehn Injection Rate Flow Is Enough To Run Calcs
460188	Wellhead #8 Flow Running Status Percent	What Percent of Desired Flow To Say Well Is Flowing For Status
460190	Wellhead #9 PID Min Output	When Running But Injection Is Less Then Flow Release
460192	Wellhead #9 PID Flow Release	Wehn Injection Rate Flow Is Enough To Run Calcs
460194	Wellhead #9 Flow Running Status Percent	What Percent of Desired Flow To Say Well Is Flowing For Status
460196	Wellhead #10 PID Min Output	When Running But Injection Is Less Then Flow Release
460198	Wellhead #10 PID Flow Release	Wehn Injection Rate Flow Is Enough To Run Calcs
460200	Wellhead #10 Flow Running Status Percent	What Percent of Desired Flow To Say Well Is Flowing For Status
460202	Reserved For Future	
460204	Reserved For Future	
460206	Reserved For Future	
460208	Wellhead Scada Epoch	
460210	Customer PLC Watchdog From Customer PLC	
460212	Wellhead #1 Injection Flow Rate From Customer PLC	
460214	Wellhead #1 Injection Static Pressure From Customer PLC	
460216	Wellhead #1 Injection Differential Prs From Customer PLC	
460218	Wellhead #1 Injection Temp From Customer PLC	
460220	Wellhead #1 Setpoint From Customer PLC	
460222	Wellhead #1 Yesterdays Total Flow	
460224	Reserved For Future	
460226	Wellhead #2 Injection Flow Rate From Customer PLC	
460228	Wellhead #2 Injection Static Pressure From Customer PLC	
460230	Wellhead #2 Injection Differential Prs From Customer PLC	
460232	Wellhead #2 Injection Temp From Customer PLC	
460234	Wellhead #2 Setpoint From Customer PLC	
460236	Wellhead #2 Yesterdays Total Flow	
460238	Reserved For Future	

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0

Page 41 of 48

460240	Wellhead #3 Injection Flow Rate From Customer PLC	
460242	Wellhead #3 Injection Static Pressure From Customer PLC	
460244	Wellhead #3 Injection Differential Prs From Customer PLC	
460246	Wellhead #3 Injection Temp From Customer PLC	
460248	Wellhead #3 Setpoint From Customer PLC	
460250	Wellhead #3 Yesterdays Total Flow	
460252	Reserved For Future	
460254	Wellhead #4 Injection Flow Rate From Customer PLC	
460256	Wellhead #4 Injection Static Pressure From Customer PLC	
460258	Wellhead #4 Injection Differential Prs From Customer PLC	
460260	Wellhead #4 Injection Temp From Customer PLC	
460262	Wellhead #4 Setpoint From Customer PLC	
460264	Wellhead #4 Yesterdays Total Flow	
460266	Reserved For Future	
460268	Wellhead #5 Injection Flow Rate From Customer PLC	
460270	Wellhead #5 Injection Static Pressure From Customer PLC	
460272	Wellhead #5 Injection Differential Prs From Customer PLC	
460274	Wellhead #5 Injection Temp From Customer PLC	
460276	Wellhead #5 Setpoint From Customer PLC	
460278	Wellhead #5 Yesterdays Total Flow	
460280	Reserved For Future	
460282	Wellhead #6 Injection Flow Rate From Customer PLC	
460284	Wellhead #6 Injection Static Pressure From Customer PLC	
460286	Wellhead #6 Injection Differential Prs From Customer PLC	
460288	Wellhead #6 Injection Temp From Customer PLC	
460290	Wellhead #6 Setpoint From Customer PLC	
460292	Wellhead #6 Yesterdays Total Flow	
460294	Reserved For Future	

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0

Page 42 of 48

460296	Wellhead #7 Injection Flow Rate From Customer PLC	
460298	Wellhead #7 Injection Static Pressure From Customer PLC	
460300	Wellhead #7 Injection Differential Prs From Customer PLC	
460302	Wellhead #7 Injection Temp From Customer PLC	
460304	Wellhead #7 Setpoint From Customer PLC	
460306	Wellhead #7 Yesterdays Total Flow	
460308	Reserved For Future	
460310	Wellhead #8 Injection Flow Rate From Customer PLC	
460312	Wellhead #8 Injection Static Pressure From Customer PLC	
460314	Wellhead #8 Injection Differential Prs From Customer PLC	
460316	Wellhead #8 Injection Temp From Customer PLC	
460318	Wellhead #8 Setpoint From Customer PLC	
460320	Wellhead #8 Yesterdays Total Flow	
460322	Reserved For Future	
460324	Wellhead #9 Injection Flow Rate From Customer PLC	
460326	Wellhead #9 Injection Static Pressure From Customer PLC	
460328	Wellhead #9 Injection Differential Prs From Customer PLC	
460330	Wellhead #9 Injection Temp From Customer PLC	
460332	Wellhead #9 Setpoint From Customer PLC	
460334	Wellhead #9 Yesterdays Total Flow	
460336	Reserved For Future	
460338	Wellhead #10 Injection Flow Rate From Customer PLC	
460340	Wellhead #10 Injection Static Pressure From Customer PLC	
460342	Wellhead #10 Injection Differential Prs From Customer PLC	
460344	Wellhead #10 Injection Temp From Customer PLC	
460346	Wellhead #10 Setpoint From Customer PLC	
460348	Wellhead #10 Yesterdays Total Flow	
460350	Reserved For Future	
460352	Reserved For Future	
460354	Reserved For Future	

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0

Page 43 of 48

460356	Reserved For Future	
460358	Reserved For Future	
460360	Compressor 1 epoch	
460362	Compressor 1 Panel State	
460364	FUXA_Compressor 1 Current_Flow	
460366	Compressor 1 Auto_Flow_Rate_PID_SP	
460368	Compressor 1 Desired_Flow_Rate_PID_SP	
460370	Compressor 1 Suction_Prs	
460372	Compressor 1 Discharge_Prs	
460374	Compressor 1 Speed_Discharge_SP	
460376	Compressor 1 Speed_Suction_SP	
460378	Reserved For Future	
460380	Compressor 2 epoch	
460382	Compressor 2 Panel_State	
460384	FUXA_Compressor 2 Current_Flow	
460386	Compressor 2 Auto_Flow_Rate_PID_SP	
460388	Compressor 2 Desired_Flow_Rate_PID_SP	
460390	Compressor 2 Suction_Prs	
460392	Compressor 2 Discharge_Prs	
460394	Compressor 2 Speed_Discharge_SP	
460396	Compressor 2 Speed_Suction_SP	
460398	Reserved For Future	
460400	Compressor 3 epoch	
460402	Compressor 3 Panel_State	
460404	FUXA_Compressor 3 Current_Flow	
460406	Compressor 3 Auto_Flow_Rate_PID_SP	
460408	Compressor 3 Desired_Flow_Rate_PID_SP	
460410	Compressor 3 Suction_Prs	
460412	Compressor 3 Discharge_Prs	
460414	Compressor 3 Speed_Discharge_SP	
460416	Compressor 3 Speed_Suction_SP	
460418	Reserved For Future	
460420	Compressor 4 epoch	
460422	Compressor 4 Panel_State	
460424	FUXA_Compressor 4 Current_Flow	
460426	Compressor 4 Auto_Flow_Rate_PID_SP	
460428	Compressor 4 Desired_Flow_Rate_PID_SP	
460430	Compressor 4 Suction_Prs	
460432	Compressor 4 Discharge_Prs	
460434	Compressor 4 Speed_Discharge_SP	
460436	Compressor 4 Speed_Suction_SP	
460438		

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0

Page 44 of 48

460440		
460442		
460444		
460446		
460448		
460450	Wellhead PLC Comms Loss	0 = OKAY, 1 = Fault
460452	Compressor 1 Comms Loss	0 = OKAY, 1 = Fault
460454	Compressor 2 Comms Loss	0 = OKAY, 1 = Fault
460456	Compressor 3 Comms Loss	0 = OKAY, 1 = Fault
460458	Compressor 4 Comms Loss	0 = OKAY, 1 = Fault
460460	Communication Lose Override Active	0 = OKAY, 1 = Active Comms Override
460462	Communication Lose Override Type	0 = Presets, 1 = Current Position
460464	Communication Lose Debounce Timer	
460466	Wellhead 1 Override Position	
460468	Wellhead 2 Override Position	
460470	Wellhead 3 Override Position	
460472	Wellhead 4 Override Position	
460474	Wellhead 5 Override Position	
460476	Wellhead 6 Override Position	
460478	Wellhead 7 Override Position	
460480	Wellhead 8 Override Position	
460482	Wellhead 9 Override Position	
460484	Wellhead 10 Override Position	

Register	Description(FROM FUXA / Mlink TO DE)	Description
461000	Total Number Of Wellheads	Value Can Be "1-10"
461002	Wellhead #1 Choke Flow Priority #	Value Can Be "1-10"
461004	Wellhead #2 Choke Flow Priority #	Value Can Be "1-10"
461006	Wellhead #3 Choke Flow Priority #	Value Can Be "1-10"
461008	Wellhead #4 Choke Flow Priority #	Value Can Be "1-10"
461010	Wellhead #5 Choke Flow Priority #	Value Can Be "1-10"
461012	Wellhead #6 Choke Flow Priority #	Value Can Be "1-10"
461014	Wellhead #7 Choke Flow Priority #	Value Can Be "1-10"
461016	Wellhead #8 Choke Flow Priority #	Value Can Be "1-10"
461018	Wellhead #9 Choke Flow Priority #	Value Can Be "1-10"
461020	Wellhead #10 Choke Flow Priority #	Value Can Be "1-10"
461022	Total Number Of Compressors	Value Can Be "1-10"
461024	Reserved For Future	
461026	Stabilization Time	0 Minutes - 6000 Minutes

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0

Page 45 of 48

461028	Staggering Delay Time	0 Minutes - 48 hours
461030	Reserved For Future	
461032	FUXA Heartbeat	FUXA to always send a value of "1"
461034	Reserved For Future	
461036	Wellhead #1 Choke Oil Priority #	Value Can Be "1-10"
461038	Wellhead #2 Choke Oil Priority #	Value Can Be "1-10"
461040	Wellhead #3 Choke Oil Priority #	Value Can Be "1-10"
461042	Wellhead #4 Choke Oil Priority #	Value Can Be "1-10"
461044	Wellhead #5 Choke Oil Priority #	Value Can Be "1-10"
461046	Wellhead #6 Choke Oil Priority #	Value Can Be "1-10"
461048	Wellhead #7 Choke Oil Priority #	Value Can Be "1-10"
461050	Wellhead #8 Choke Oil Priority #	Value Can Be "1-10"
461052	Wellhead #9 Choke Oil Priority #	Value Can Be "1-10"
461054	Wellhead #10 Choke Oil Priority #	Value Can Be "1-10"
461056	Reserved For Future	
461058	Reserved For Future	
461060	Reserved For Future	
461062	Compressor #1 Max Flow Rate MMSCFD	Default: 200hp @1.0 Million, 450HP @2.2 Million, 300hpgl @ 3.5 Million, 900hp @ 4.5 Million, Kta cummins @ 1.6 Million
461064	Compressor #2 Max Flow Rate MMSCFD	Default: 200hp @1.0 Million, 450HP @2.2 Million, 300hpgl @ 3.5 Million, 900hp @ 4.5 Million, Kta cummins @ 1.6 Million
461066	Compressor #3 Max Flow Rate MMSCFD	Default: 200hp @1.0 Million, 450HP @2.2 Million, 300hpgl @ 3.5 Million, 900hp @ 4.5 Million, Kta cummins @ 1.6 Million
461068	Compressor #4 Max Flow Rate MMSCFD	Default: 200hp @1.0 Million, 450HP @2.2 Million, 300hpgl @ 3.5 Million, 900hp @ 4.5 Million, Kta cummins @ 1.6 Million
461070	Reserved For Future	
461072	Reserved For Future	
461074	Reserved For Future	
461076	Reserved For Future	
461078	Reserved For Future	
461080	Reserved For Future	
461082	Reserved For Future	
461084	Reserved For Future	
461086	Reserved For Future	

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0

Page 46 of 48

ALTRONIC

461088	Reserved For Future	
461090	Reserved For Future	
461092	Reserved For Future	
461094	Reserved For Future	
461096	Reserved For Future	
461098	Reserved For Future	
461100	Reserved For Future	
461102	Reserved For Future	
461104	Reserved For Future	
461106	Reserved For Future	
461108	Reserved For Future	
461110	Reserved For Future	
461112	Reserved For Future	
461114	Reserved For Future	
461116	Reserved For Future	
461118	Reserved For Future	
461120	Reserved For Future	
461122	Reserved For Future	
461124	Reserved For Future	
461126	Reserved For Future	
461128	Reserved For Future	
461130	Reserved For Future	
461132	Reserved For Future	
461134	Wellhead #1 Max Flow Rate MMSCFD	
461136	Wellhead #2 Max Flow Rate MMSCFD	
461138	Wellhead #3 Max Flow Rate MMSCFD	
461140	Wellhead #4 Max Flow Rate MMSCFD	
461142	Wellhead #5 Max Flow Rate MMSCFD	
461144	Wellhead #6 Max Flow Rate MMSCFD	
461146	Wellhead #7 Max Flow Rate MMSCFD	
461148	Wellhead #8 Max Flow Rate MMSCFD	
461150	Wellhead #9 Max Flow Rate MMSCFD	
461152	Wellhead #10 Max Flow Rate MMSCFD	

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0

Page 47 of 48

©2025 Altronic, LLC

Sequence of operations Altronic Control Panel Wellhead 2.0

Page 48 of 48

ALTRONIC